NLP Assignment 1

Q1

We expect the brown trigram model to show lower perplexity than the brown bigram model as the trigram model would have more information on the context of the sentence. This would indicate that the trigram model should provide more coherent sentences.

But when testing brown_bigrams and brown_trigrams, we see that the mean perplexity is lower for bigrams.

This is not expected. The reason we see this might be because we are only assessing our model only with a few sentences and results might hence vary.

The code snippets are given below:

```
perp_values=[]
for i in range(5):
    perp_values.append(perplexity(brown_corpus[i], bigrams_brown, all_words))

print(sum(perp_values)/len(perp_values))
```

```
2105.2210692456542
```

```
perp_values=[]
for i in range(5):
    perp_values.append(perplexity_trigrams(brown_corpus[i], trigrams_brown, bigrams_brown, all_words))

print(sum(perp_values)/len(perp_values))
```

```
4414.472328046814
```

Q2

First we calculate the perplexities with the brown corpus.

```python
perp_values=[]
for i in range(25):
    perp_values.append(perplexity(reuters_corpus[i], bigrams_brown, all_words))

print(sum(perp_values)/len(perp_values))
```

```
5506.604062485168
```

```python
perp_values=[]
for i in range(25):
    perp_values.append(perplexity_trigrams(reuters_corpus[i], trigrams_brown, bigrams_brown, all_words))

print(sum(perp_values)/len(perp_values))
```

```
10358.815828025346
```

Now calculating the perplexities of the webtext corpus.

```python
perp_values=[]
for i in range(25):
    perp_values.append(perplexity(reuters_corpus[i], bigrams_webtext, all_words))

print(sum(perp_values)/len(perp_values))
```

```
4823.552140736586
```

```python
perp_values=[]
for i in range(25):
    perp_values.append(perplexity_trigrams(reuters_corpus[i], trigrams_webtext, bigrams_webtext, all_words))

print(sum(perp_values)/len(perp_values))
```

```
8331.805068795325
```

We can see that we get lower perplexities in case for both bigrams and trigrams with the webtext corpus. This shows us that the webtext corpus is more suited for the Reuters corpus than the brown corpus. There could be a number of reasons for this:

- The webtext corpus has more words in common with the Reuters corpus which reduces unseen words.
- The writing style in webtext corpus would be more similar to reuters than browns.

Q3

When we increase the number of sentences in our training data, we can generally expect better generalization by our model. This can be because increasing the amount of data, would result in better frequency estimates of bigrams and trigrams which would result in more accurate probabilities. Increasing the training data could also result in an improved vocabulary and would lessen the chances of unseen words. It would also reduce the chances of overfitting. And as the generalization of the model increases, we expect a lower perplexity score.

Although after a certain point, increasing the data could result in diminishing returns as it could lead to slower computation and the improvements made by the model could be very minimal.

To conclude we should expect more legible and grammatically correct sentences as our model would generalize better.